

Manuscript Supplement V1

Date: 2026-04-11

Target venue: Empirical Software Engineering

Purpose

Provide the submission-facing supplement that supports the main manuscript without forcing the main paper to carry internal run-accounting detail.

Supplement A. Experimental Conditions

Core condition definitions

- **baseline**: the weaker comparison condition that does not receive the reusable protocol artifact being tested
- **taught**: the comparison condition that receives the reusable protocol artifact directly
- **correction**: a two-model condition in which the reviewer critiques and the original executor remains final author
- **takeover**: a two-model condition in which the reviewer or second model becomes the final author
- **quality-bar loop**: a bounded revise-and-check loop with explicit pass and stop conditions

Status labels used in workflow branches

- **pass-core**: the minimum success bar for a workflow artifact chain
- **pass-stretch**: a stronger-than-required success outcome
- **fail-cap**: the loop cap was reached without clearing the required bar
- **blocked**: the run could not proceed cleanly because of a method, runtime, or account-side blocker

Construct note

In this manuscript, **teaching** does not mean durable internal learning. It means that a lesson has been externalized into a reusable protocol artifact and then injected into the taught condition as a structured intervention.

Supplement B. Branch-Level Run Accounting

Hosted manuscript spine

The hosted manuscript spine is built from OpenAI gpt-4.1, Google gemini-2.5-flash, and Anthropic claude-sonnet-4-6 across:

- BP001 planning
- BP002 structured workflow

- BP003 harder workflow and transfer
- BP004 higher-realism proxy-materialized workflow

The main hosted synthesis uses the current retained comparison sets recorded in:

- docs/benchmarks/cross-provider-synthesis-2026-03-13.md

Open-weight `BP001`

Current branch-level universe:

- five local models
- one baseline and one taught condition per model
- ten scored condition outputs total

Models in the frozen comparison surface:

- qwen2.5:3b
- qwen2.5-coder:3b
- llama3.1:8b
- llama3.2:3b
- deepseek-coder-v2:16b-lite-instruct-q3_K_S

Primary ledger:

- docs/benchmarks/open-weight-run-sheet-v1.md

`BP006` mixture-of-experts branch

Current branch-level universe:

- five eligible models
- twenty ordered executor-reviewer pairs excluding self-review
- six retained and scored pairs
- fourteen planned pairs at the current snapshot

Primary ledger:

- docs/benchmarks/bp006-exhaustive-run-sheet-v1.md

Interpretation rule:

- report BP006 as structured exploratory architecture evidence, not as an exhaustive completed matrix

`BP007B` verifier-backed local planning branch

Current branch-level universe:

- indexed slots T01 through T09

- nine total branch slots

Primary ledger:

- docs/benchmarks/bp007b-run-sheet-v1.md

Interpretation rule:

- methodologically useful diagnostic branch, but not the main local publication surface

`BP008` local workflow branch

Current branch-level universe:

- W01
- W02
- W03

Primary ledger:

- docs/benchmarks/bp008-run-sheet-v1.md

Interpretation rule:

- strongest current local resilience surface

Supplement C. Retained And Superseded Logic

Working definition

Retained means:

- preserved as the current evidence-bearing artifact chain for a run, pair, or branch slot
- not deleted or replaced in the audit trail without documentation
- available for synthesis, review, or manuscript reference

Retained does not mean:

- the only run ever attempted
- the visually strongest result chosen after broad hidden exploration
- the full run universe for a branch

Main retention cases

First stable completed run

Some branches retain the first cleanly completed run as the current evidence-bearing artifact chain.

Clarified rerun replacing an ambiguous earlier run

Some branches required a rerun because an earlier run was not methodologically clean enough for publication use, for example due to:

- packet ambiguity
- clarified placeholder policy
- tightened reviewer logic
- runtime invalidity

In those cases, the earlier run remains in the audit trail and the clarified rerun becomes the current evidence-bearing point.

Exploratory subset

Some branches, especially BP006, use a retained subset for synthesis even though the broader branch is incomplete.

Reporting rule:

- say explicitly that the branch is exploratory or partially complete
- do not let `retained` stand in for exhaustive accounting

Supplement D. Scoring And Review Procedures

Review layers

- L0: initial operator scoring against the benchmark rubric
- L1: operator-side second-pass review or adjudication where applicable
- L2A: model-based secondary review, currently Anthropic on the hosted BP001 and BP002 slice
- L2: distinct human evaluator review, currently completed for the hosted BP001 and BP002 six-run slice

Current review posture

- most of the full portfolio still relies on L0/L1
- L2A is used as a secondary consistency check, not as independence proof
- distinct human L2 now exists for the hosted BP001/BP002 slice and agrees directionally with the stronger taught condition in all six reviewed runs

Human `L2` handling rule

The Fernanda De La O memo is the controlling human-review artifact. The manuscript uses the memo for directional support and uses the repository-local score-table transcription only for column totals. The paper does not rely on unlabeled rubric rows for finer-grained numerical interpretation.

Supplement E. Results Tables

Hosted cross-provider within-model uplift

Provider	Model	BP001 baseline	BP001 taught	BP002 baseline	BP002 taught	BP003 baseline	BP003 taught
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Provider	Model	BP001 baseline	BP001 taught	BP002 baseline	BP002 taught	BP003 baseline	BP003 taught
OpenAI	gpt-4.1	3/10	9/10	5/10	9/10	8/10	10/10
Google	gemini-2.5-flash	3/10	7/10	8/10	9/10	8/10	10/10
Anthropic	claude-sonnet-4-6	1/10	8/10	8/10	9/10	8/10	10/10

Local evidence ladder

Surface	Strongest retained result	Interpretation
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Open-weight BP001	taught ceiling 6/10	structural planning uplift exists, but absolute planning fidelity remains limited
BP007B verifier-backed planning	T07 reaches pass-core without score uplift	stronger verification removes false positives but does not rescue weak local planning
BP008 workflow branch	W01 and W03 reach pass-core	concrete workflow tasks preserve the clearest local resilience effect

`BP006` architecture summary

Pair	Executor	Reviewer	Best condition
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P1	gpt-4.1	claude-sonnet-4-6	correction
P2	gemini-2.5-flash	gpt-4.1	takeover
P3	gpt-4.1	gemini-2.5-flash	no material MoE gain
P4	claude-sonnet-4-6	gpt-4.1	correction
P5	deepseek-coder-v2:16b-lite-instruct-q3_K_S	gpt-4.1	takeover
P6	deepseek-coder-v2:16b-lite-instruct-q3_K_S	llama3.2:3b	modest local-local gain

Supplement F. Local Runtime And Environment Constraints

The local branch should be interpreted with the following environmental constraints in mind:

- one-model-at-a-time storage discipline on the current machine
- slower local inference than hosted runs
- lower-quality all-local review loops than hosted-review loops
- stronger sensitivity to runtime instability during revise-and-check cycles

These constraints matter because local ceilings reflect both model capability and operating environment. The local branch therefore supports resilience claims, not hosted-quality parity claims.

Supplement G. Data, Materials, And Audit Trail Availability

The following materials are archived in the project repository and available as part of the manuscript evidence chain:

- benchmark packets
- run sheets

- retained artifact chains
- synthesis memos
- scoring rubrics
- human and model-based hosted review materials

The separate EMSE title-page declarations currently state that the archived materials are maintained in a project repository that is not presently released as a public dataset because it includes operational materials and review artifacts, but editor and reviewer access can be provided on reasonable request.